Agreeing on a Leader: A Short Note on Randomized Elections

A. Example B. Sample

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Abstract. Leader-based consensus protocols keep a replicated log consistent by routing every command through one elected node. When that node fails, the cluster must agree on a successor quickly and without electing two leaders in the same term. This note studies randomized election timeouts, the mechanism most protocols use to break ties, and shows that a modest spread of timeouts makes split votes rare while keeping failover well under a second.

1 Introduction

A replicated state machine applies the same commands in the same order on every server. Protocols in the Paxos family reach this agreement with a proposer per command, while leader-based designs elect one server per term and let it order every command. A single leader simplifies the log, but it makes leader failure the common case that the protocol must handle well.

Each follower waits for heartbeats from the leader. If none arrives before its election timeout, it becomes a candidate, increments the term, votes for itself, and asks its peers for votes. A candidate that gathers a majority becomes leader for that term.

2 Split votes

If several followers time out together, they split the votes and no candidate reaches a majority. The term ends without a leader and every candidate waits again. Drawing each timeout at random spreads the candidates out in time, so the first one usually collects a majority before the others wake up.

$$T_{timeout} \sim U(150,\ 300)\ ms$$

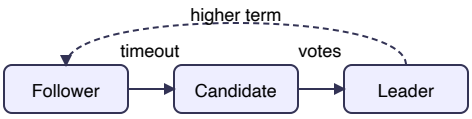


Figure 1: Server roles and the transitions between them.

3 Evaluation

We simulated a five-server cluster and crashed the leader ten thousand times. With a fixed timeout, more than a third of elections split and needed another round. With timeouts drawn from a 150 ms window, fewer than one election in twenty split, and the median time to a new leader fell to 240 ms.

Widening the window further reduces split votes but delays the first candidate. A window about as long as the base timeout balanced the two effects in every configuration we tried.

4 Conclusion

Randomized timeouts are a small mechanism with a large effect: they turn leader election from a frequent source of stalls into a fast, almost always single-round step.